

Body-fitted NACA0012 verification

Abstract

Three actual polygon-grid simulations assess steady laminar external flow. Numerical convergence and strict physical error below 3% are separate gates. Complete raw fields, force components, histories, independent NumPy equation audits and source hashes are supplied. Failed metrics remain failed.

1. Physical problem

NACA0012: $Re=1000$, angle of attack 4 degrees, unit chord, domain 20×16 , leading edge (6,8). Laminar, incompressible, steady, two-dimensional flow. The public reference is Di Ilio et al. (2020), arXiv:2006.10487, figures 10 and 11.

2. Method and mesh

Collocated SIMPLE uses first-order upwind momentum transport, nonorthogonal diffusion and Rhie-Chow face flux. Walls are polygon chords. CPU SuperLU solves the same finite-volume linear equations with true residual checks. The original SIMPLE equations are retained; the finest grid uses current-matrix GMRES with cached LU preconditioning and existing Anderson mixing. Post-mixing residuals and physical flux are explicitly checked. The radial periodic airfoil grid keeps its c-grid API name. Coupled Picard acceleration is not certified for airfoils and is rejected by the production interface.

3. Acceptance and reproducibility

Each integrated physical reference error must be strictly below 3%; steady momentum, continuity and boundary imbalance must be below $1e-6$. Independent NumPy gradients, complete momentum residuals, mass and inertia-free physical Rhie-Chow flux are audited. Grid refinement alone does not prove an asymptotic regime. Exact commands and editable LaTeX source accompany this PDF. Elapsed solve time excludes mesh setup, export and plotting; no matched TensorLBM performance comparison is claimed.

4. Reference integrity

Vector-digitized present-study markers give $Cl=0.2009636030$ and $Cd=0.1250065462$. Each has a conservative graph-reading interval ± 0.0001 . The worst relative error at either endpoint is used. This interval excludes author numerical uncertainty. The historical 0.205/0.120 pass claim is withdrawn. No matching four-degree Cp reference is claimed.

5. Results and independent audits

CPU float64; three actual grids. Physical errors are evaluated separately.

Grid	Cells	Steps	Steady	Cd	Cl	Max physical error (%)	Qualified
64x26	1664	608	True	0.1209792	0.2062255	3.299	False
128x52	6656	915	True	0.1487814	0.2429964	20.976	False
256x104	26624	546	True	0.1448016	0.2329089	15.954	False

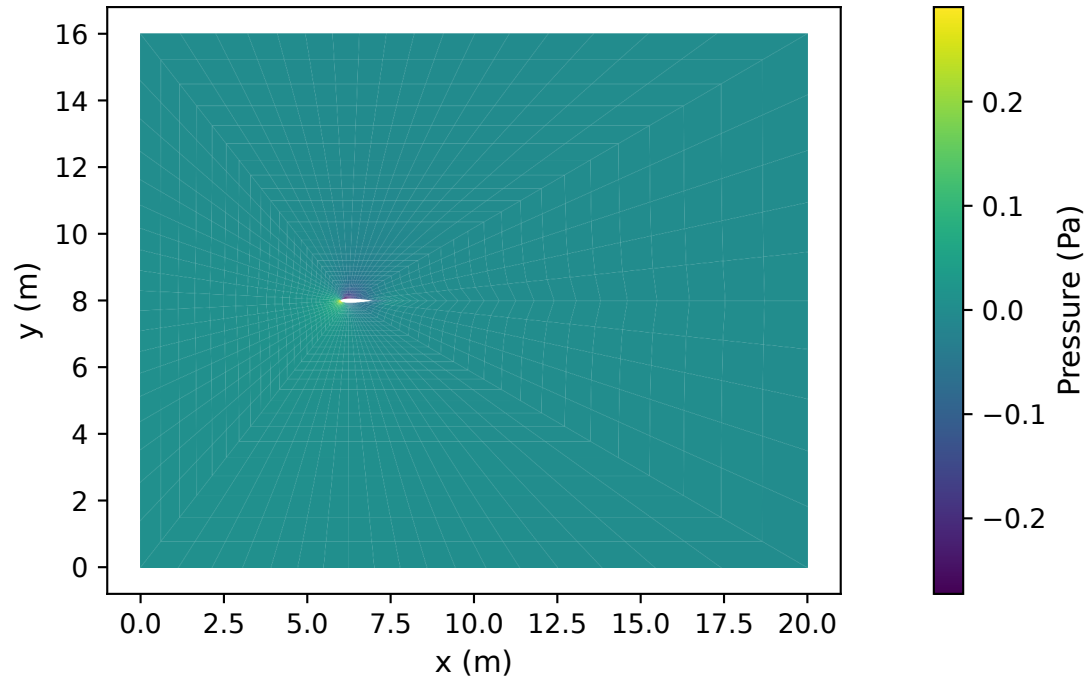
Independent equations reconstructed from saved fields

Grid	Momentum	Continuity	Physical flux defect	Fixed point
64x26	9.941e-07	2.776e-17	4.223e-10	True
128x52	9.888e-07	2.082e-17	2.668e-11	True
256x104	9.998e-07	1.041e-17	1.880e-11	True

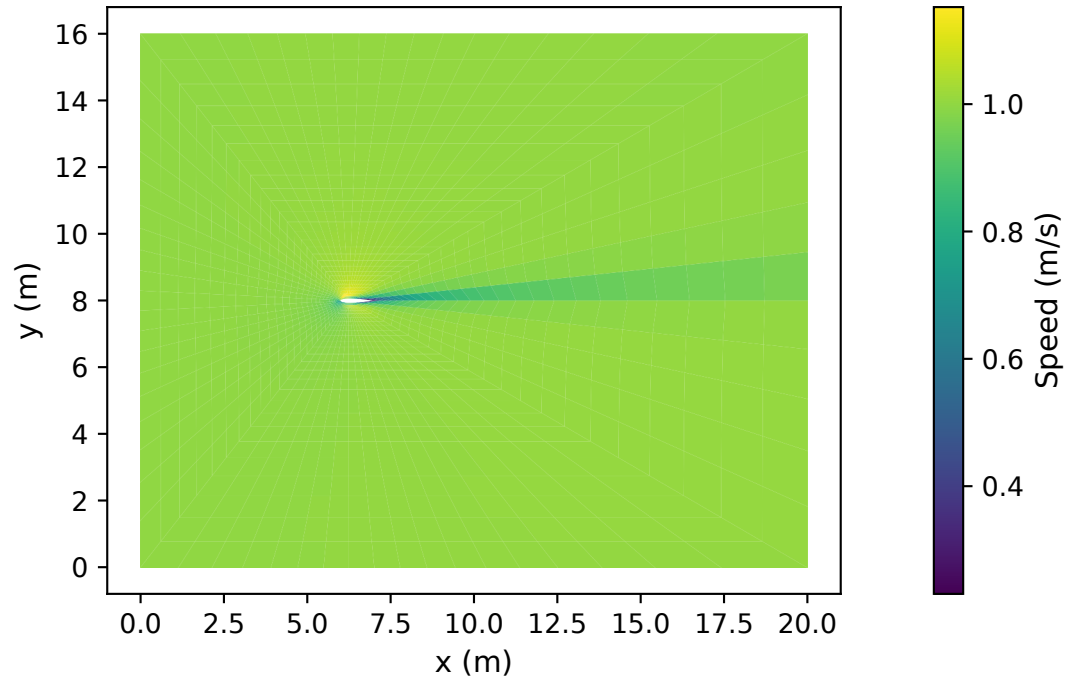
6. Discussion and limits

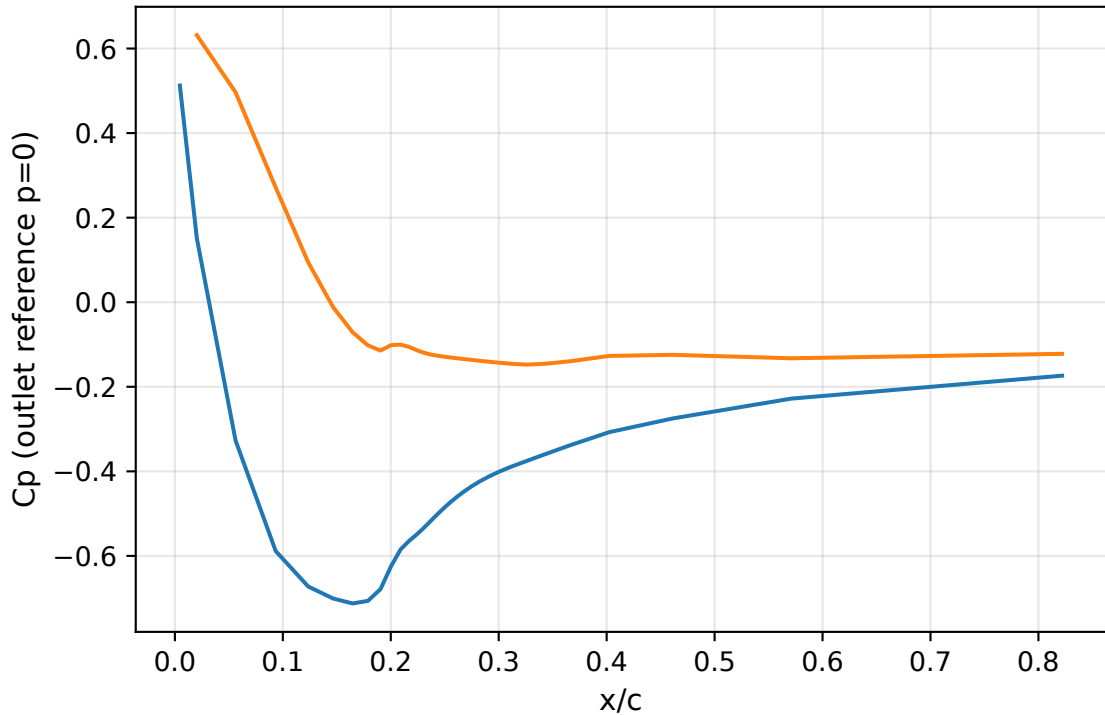
All physical gates must pass independently. Iterative convergence cannot replace physical accuracy. The finest mesh has 26624 control volumes and 79872 primary velocity/pressure unknowns. First-order transport, wall traction, polygon geometry and finite-domain effects require further separation before qualification.

naca 64x26

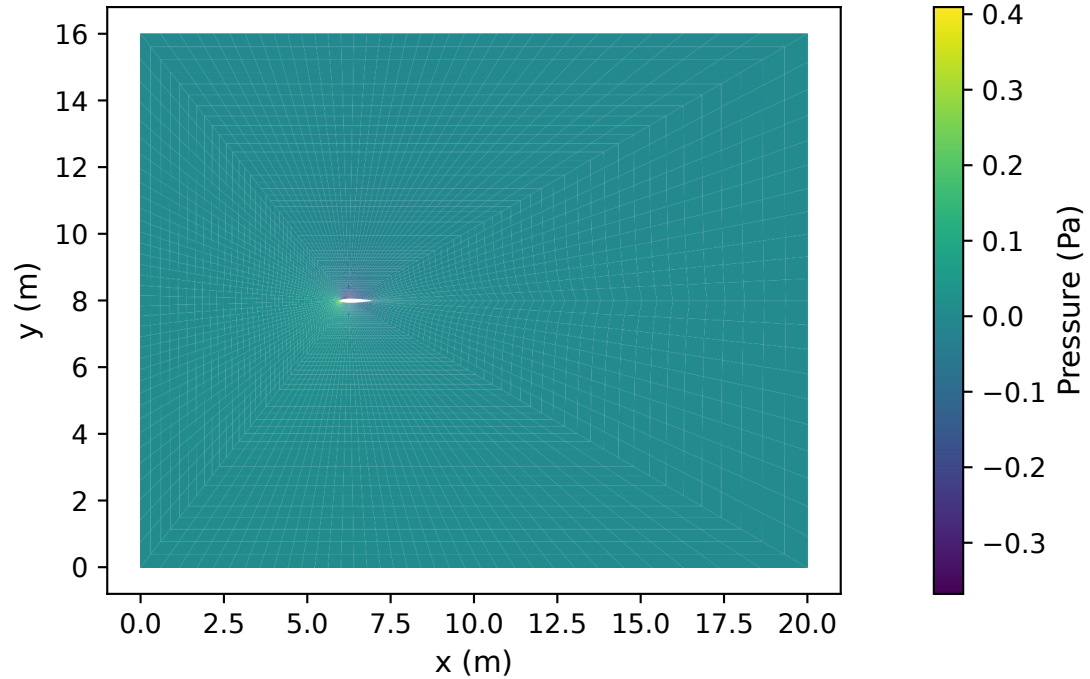


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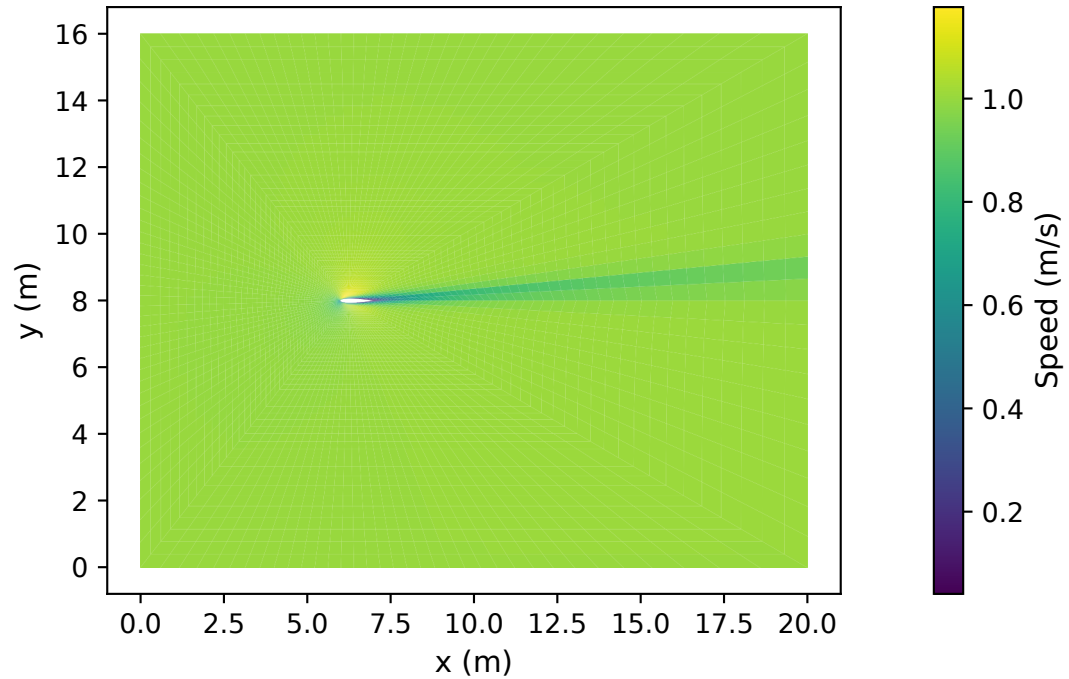


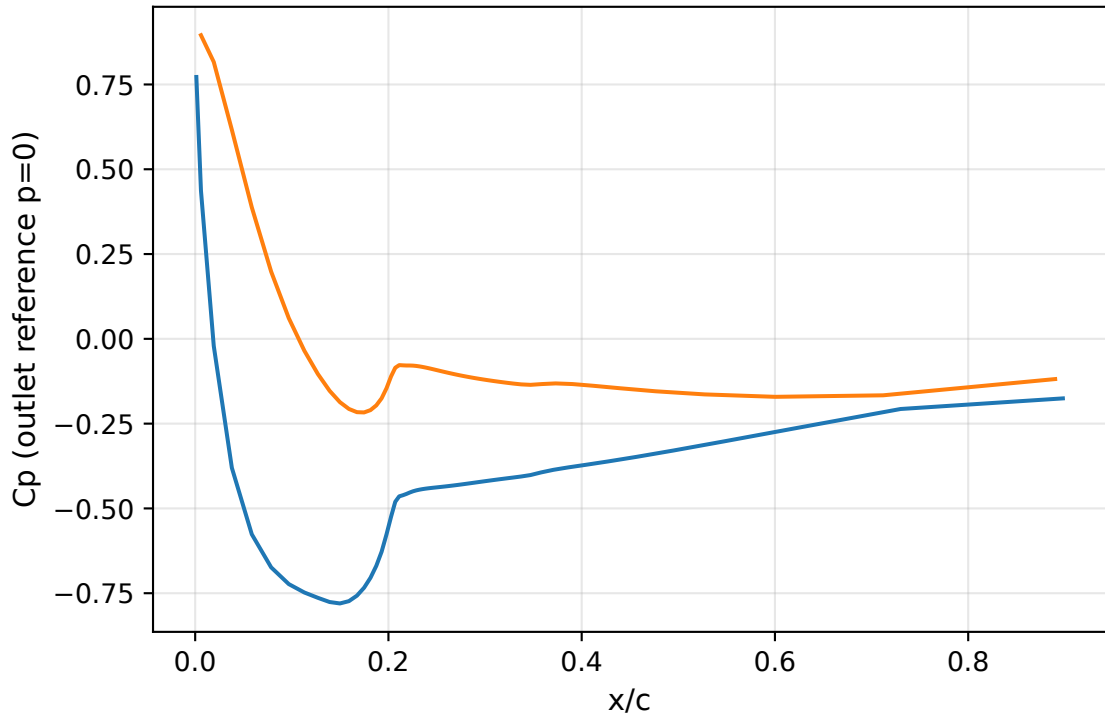


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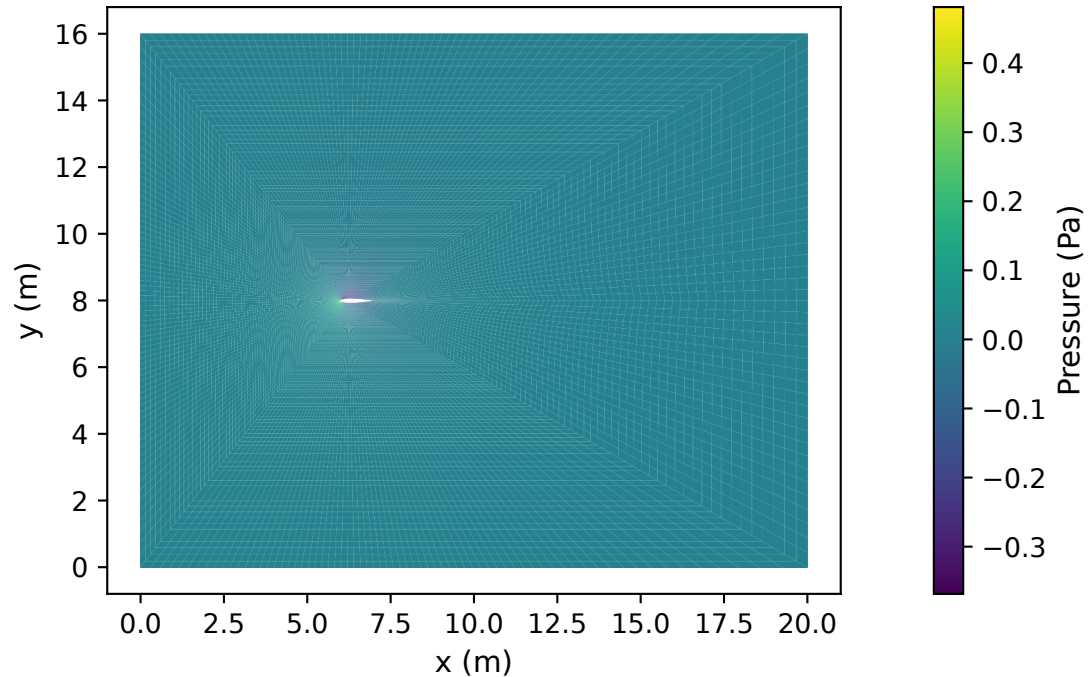


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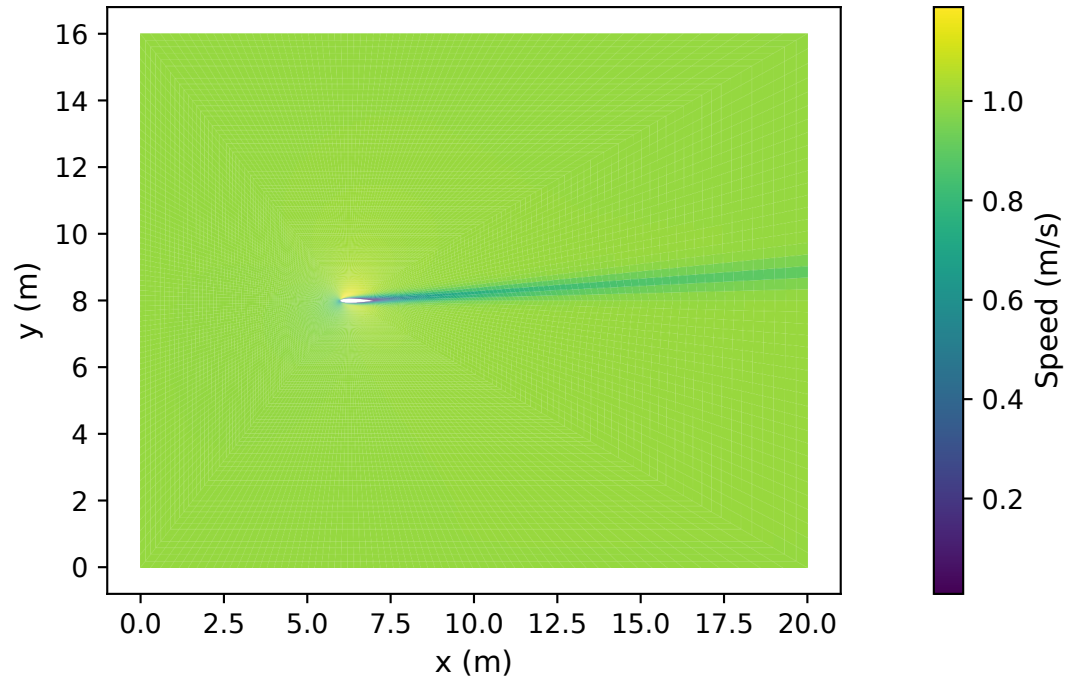


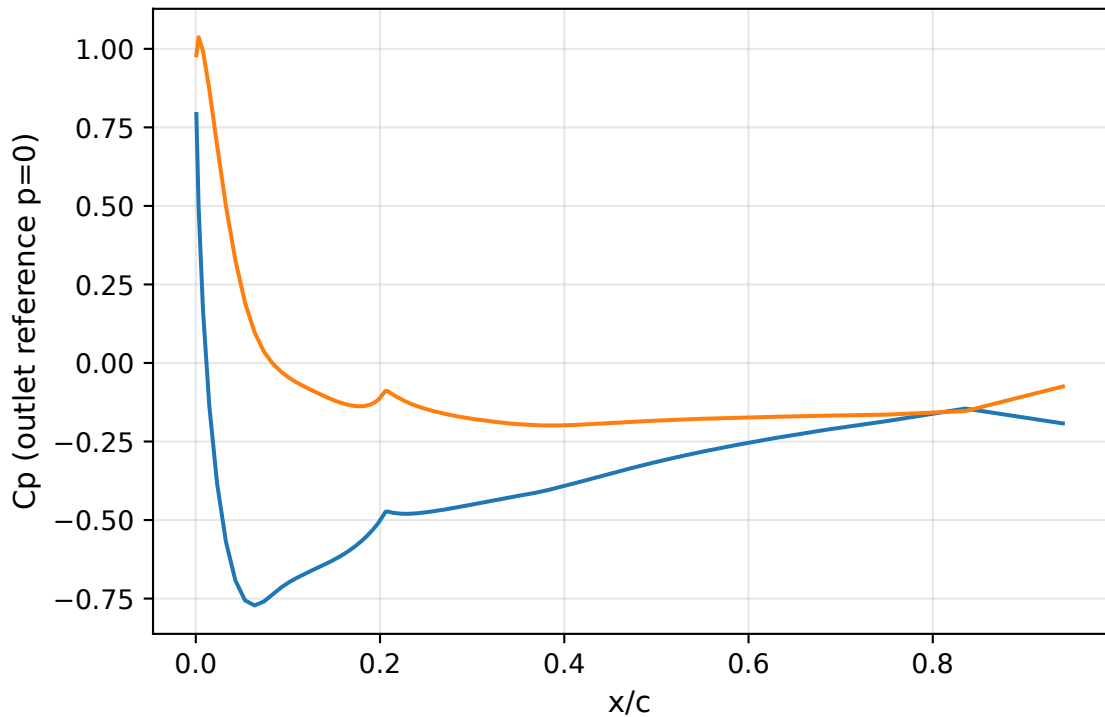


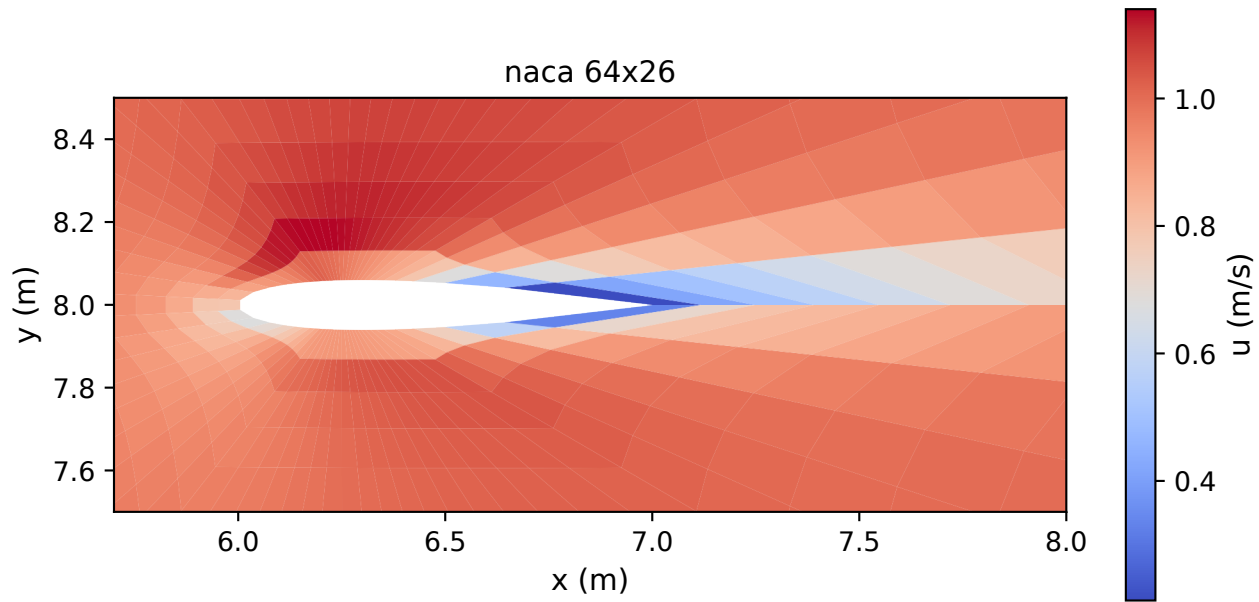
naca 256x104

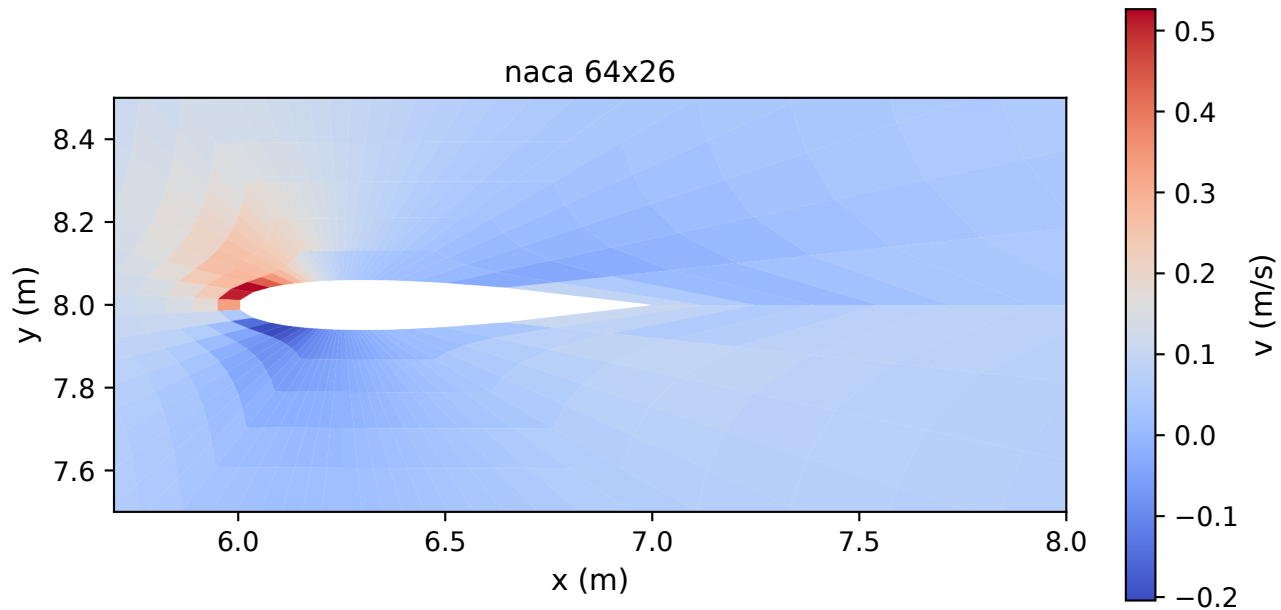


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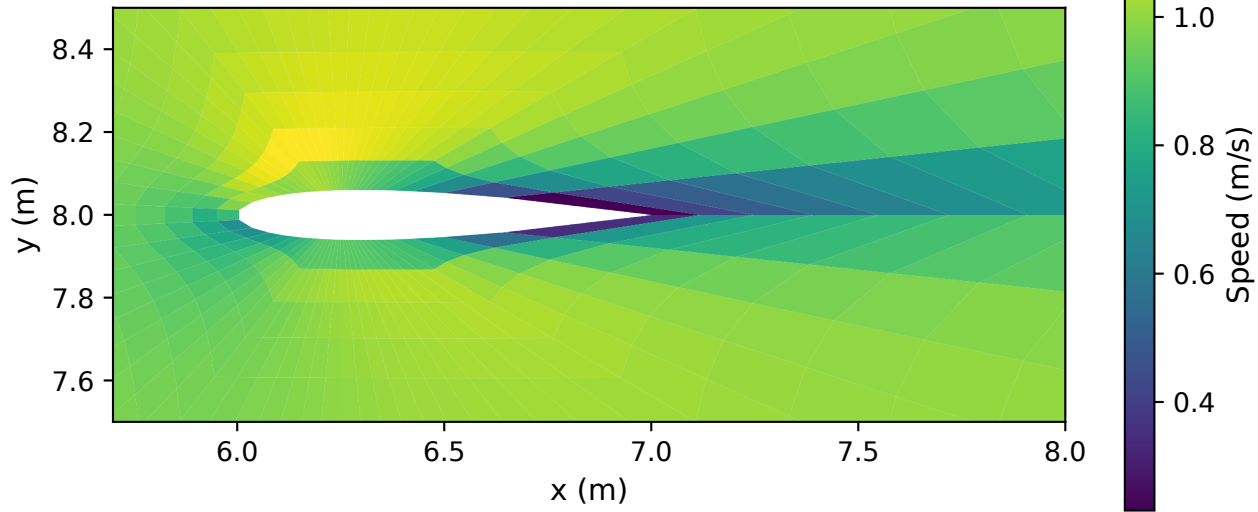




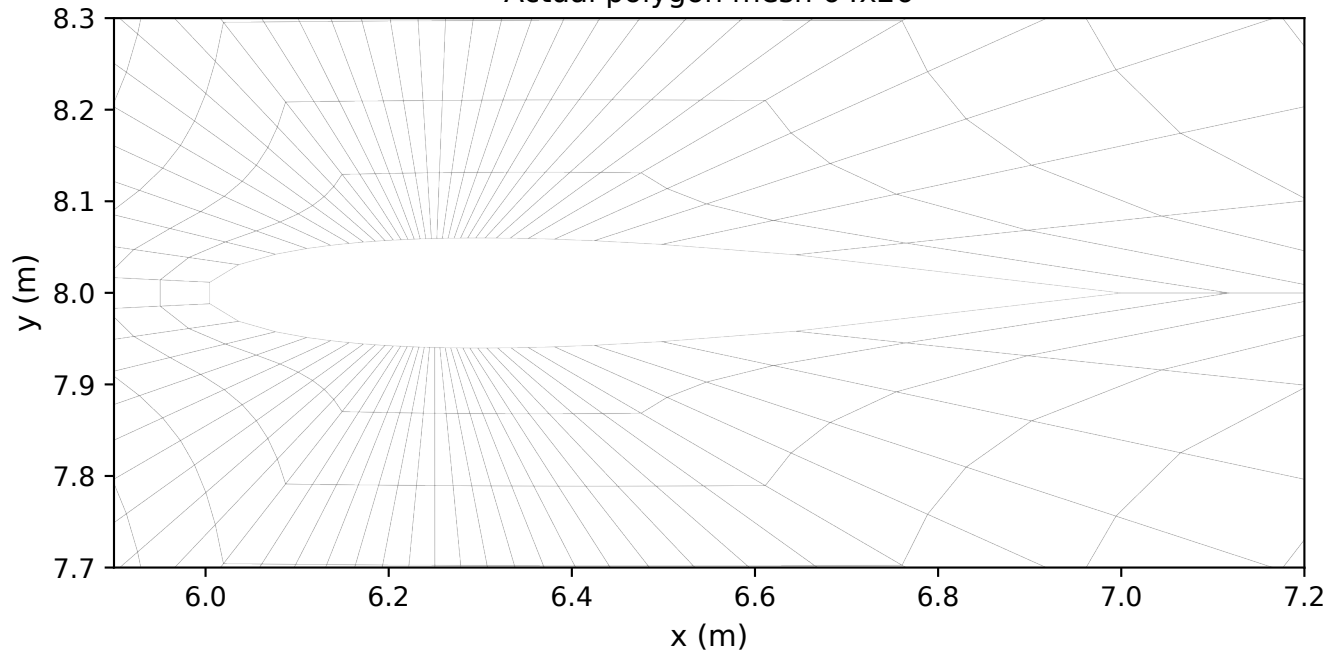


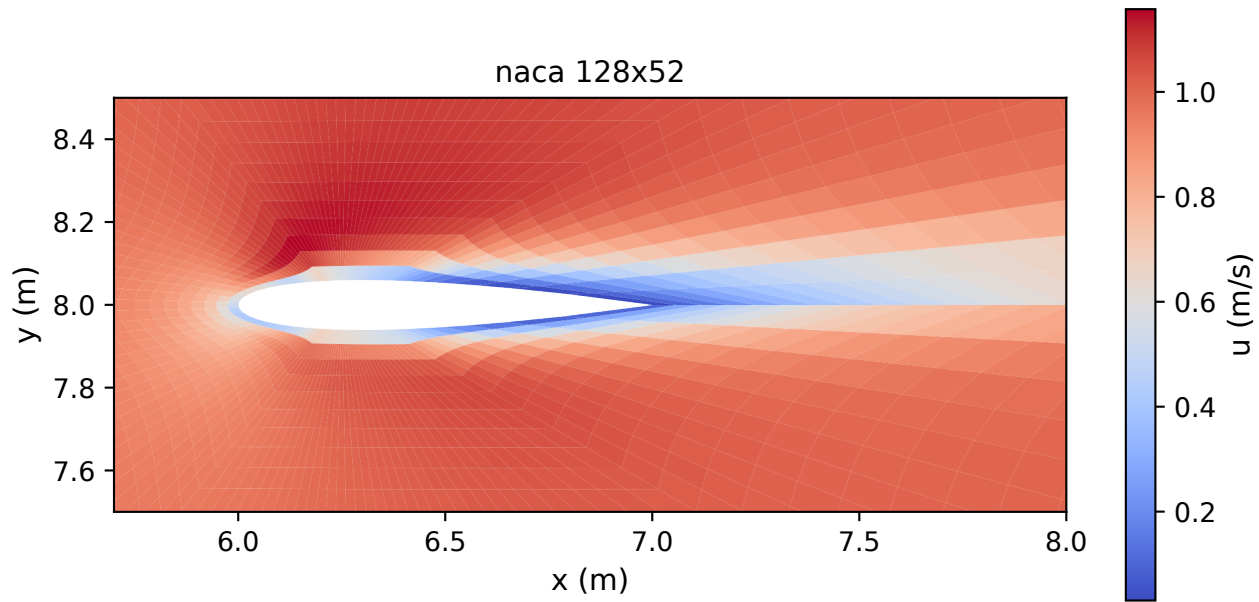


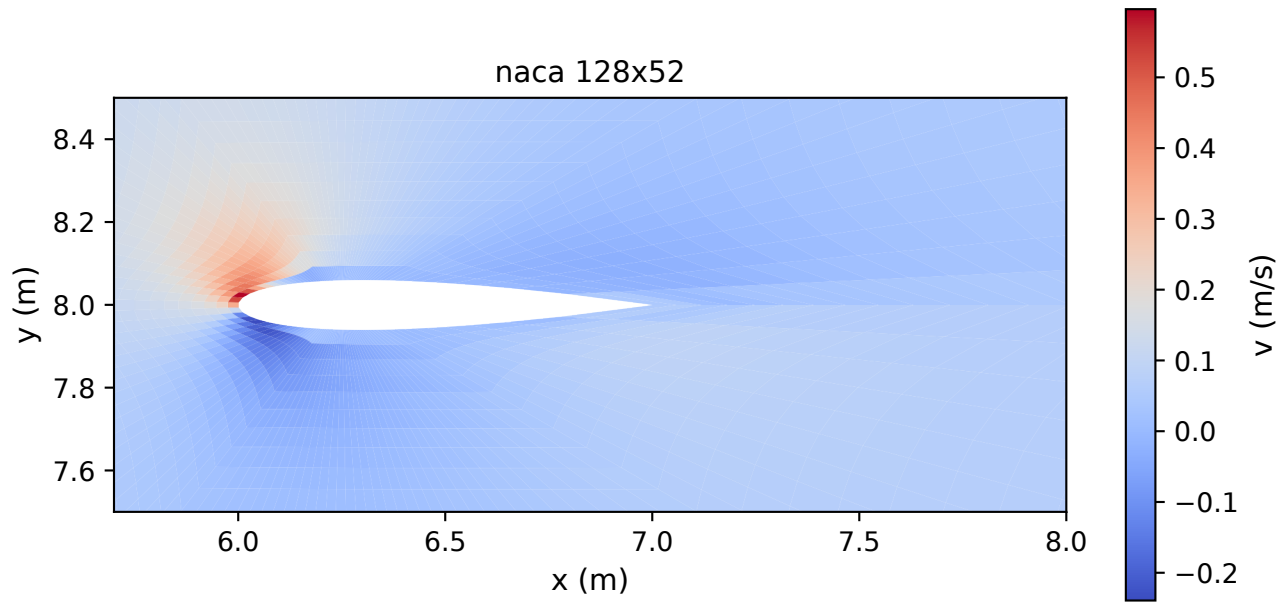
Wall and wake detail: 64x26

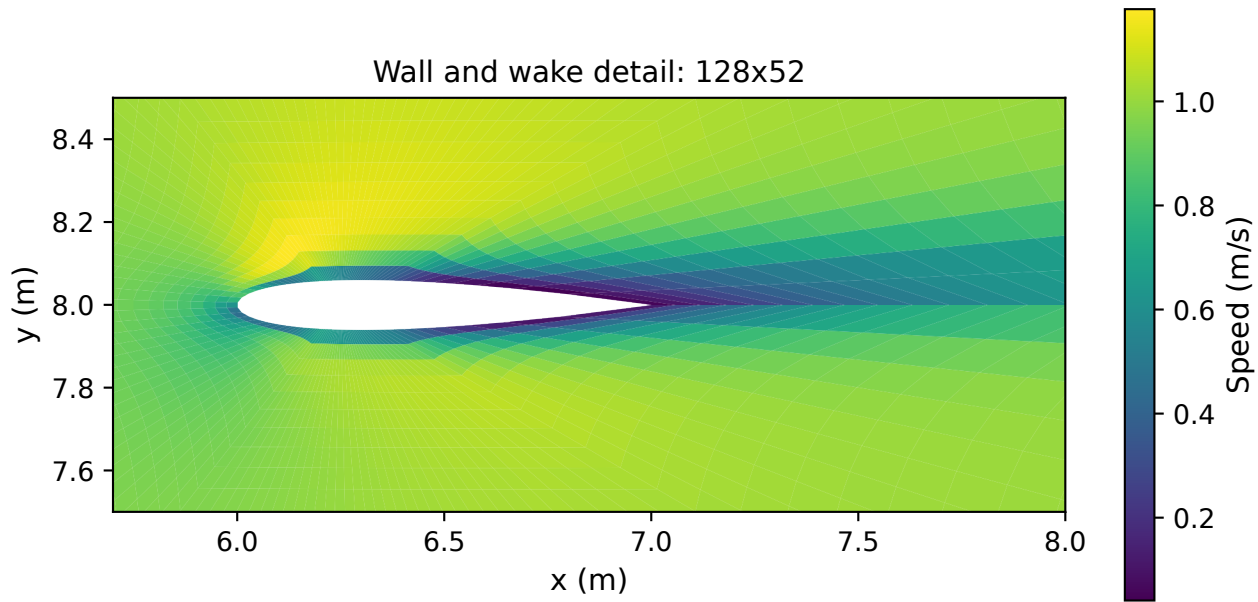


Actual polygon mesh 64x26

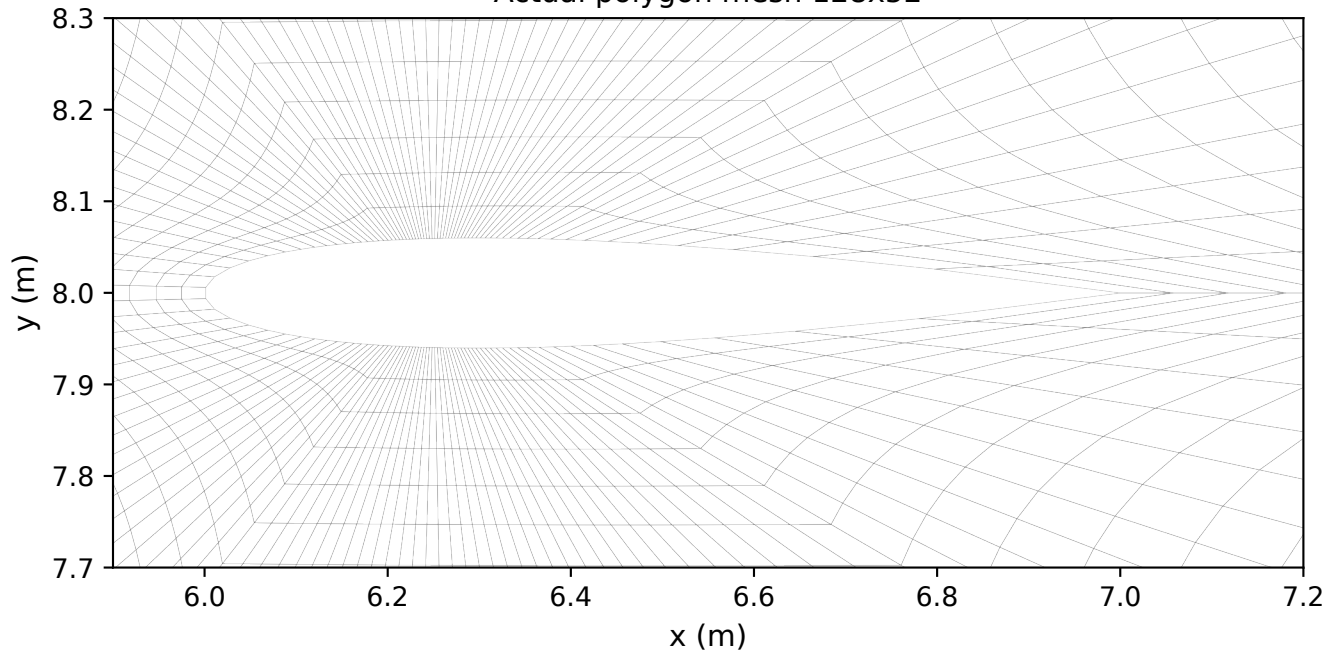


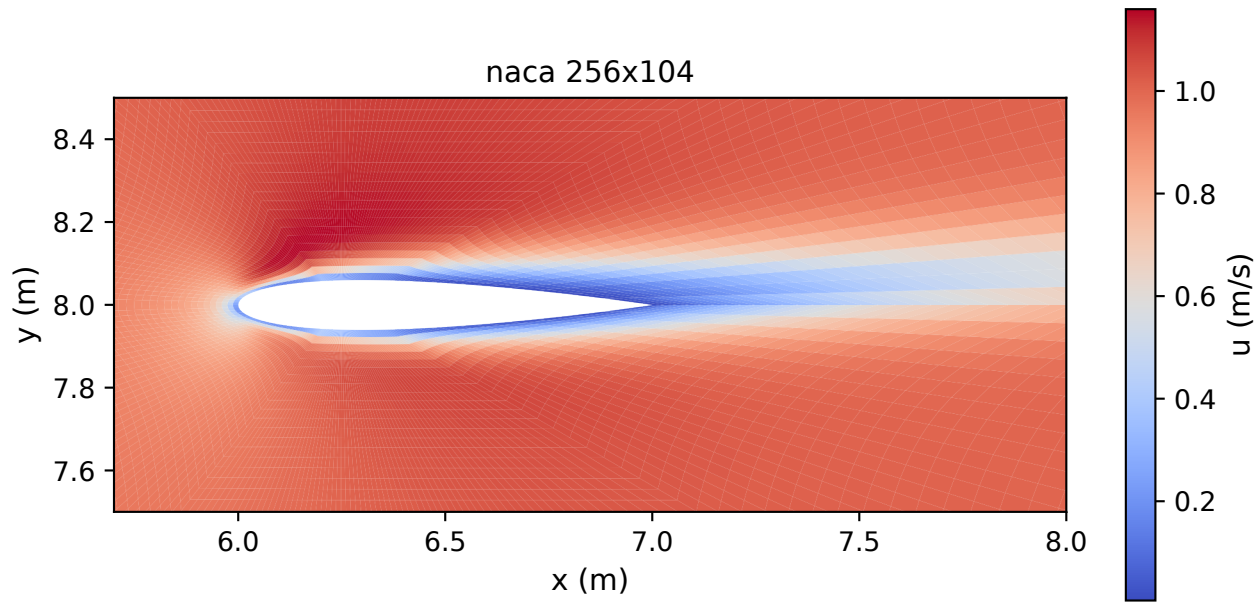


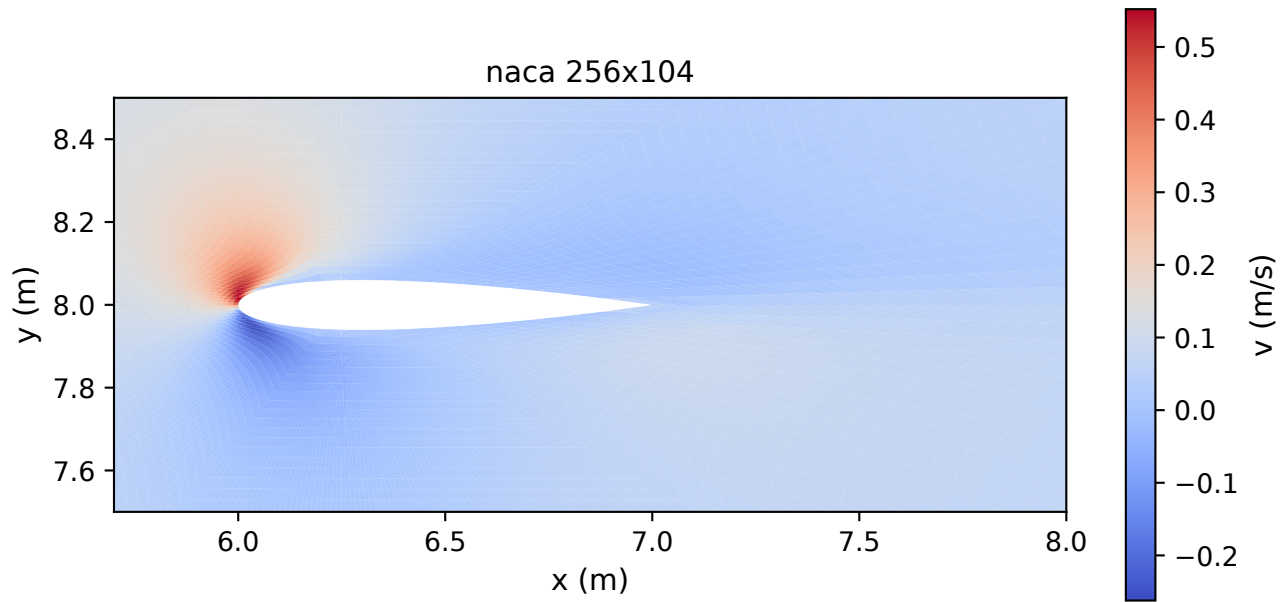




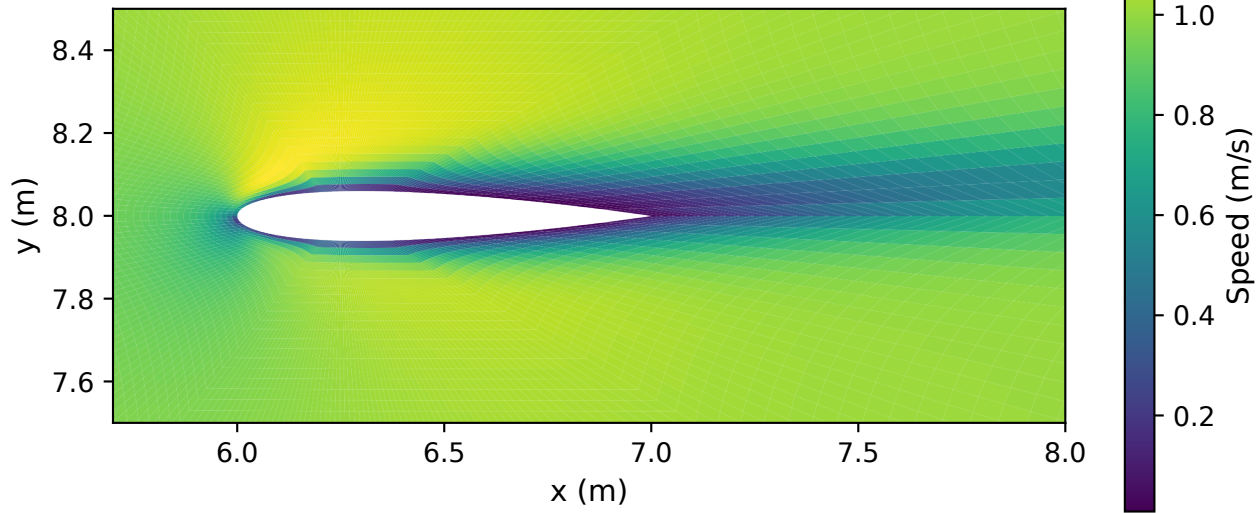
Actual polygon mesh 128x52







Wall and wake detail: 256x104



Actual polygon mesh 256x104

